



Posted Early, Swooped All the Same: A Quasi-Experimental Comparison of Anticipatory and Reactive Magpie-Warning Sign Placement

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Abstract. Facilities services that manage seasonal magpie-swooping risk face a timing choice: post warning signs ahead of a forecast return date, informed by the University's nest register and the prior season's reports, or wait and post reactively once a path's first swoop of the season is reported. The anticipatory approach carries the implicit assumption that earlier signage suppresses subsequent incidents on that path. We test the assumption directly across 46 monitored campus paths over one ten-week swooping season: 19 received anticipatory signage a mean 9.4 days ahead of the forecast return date, and 27 received signage only after the path's first reported swoop, a mean 1.8 days later. Weekly incident-report rates were statistically indistinguishable between the two groups across every phase of the season, and forecast lead-time showed no reliable association with a path's total-season incident count ($r = 0.07$, n.s.). A logistic model regressing weekly incident probability on lead-time, foot traffic, and prior-season history found the lead-time term contributed negligibly to predicted probabilities once the other two were included; we retain it for completeness. Signage timing, in other words, tracked the University's own anticipatory-capability doctrine closely and swoop-incident rates not at all. We discuss what anticipatory placement may be accomplishing instead.

Introduction

Australian magpies (*Gymnorhina tibicen*) defend nest sites against perceived intruders for a bounded window each spring, and a fraction of breeding males target pedestrians and cyclists specifically enough to generate a predictable seasonal spike in reported incidents on affected paths [1], [2]. University Facilities services have responded with warning signage, typically posted once a path's first swoop of the season is reported. In recent years, the University's nest register has supported a second, anticipatory practice: forecasting which paths are likely to return to active defence from the prior season's nest sites and swoop reports, and posting signage ahead of that forecast return date rather than waiting for the first incident.

The anticipatory practice sits squarely within the School of Emergent Priorities' broader pro-

gramme of acting on a signal before it is confirmed by direct experience, and Facilities has adopted it on the working assumption — untested until now — that earlier signage suppresses the incidents that follow. Whether a lead time of days translates into a measurably safer path, or whether swoop incidence is set by factors the sign cannot touch (bird identity, path geometry, pedestrian density), has not been examined.

This paper reports a quasi-experimental comparison of the University's anticipatory and reactive sign-placement paths across one ten-week swooping season. Our contributions are three-fold:

- we compare weekly swoop-incident-report rates between anticipatory- and reactive-signage paths across matched phases of the

season, a comparison we believe is informative regardless of its direction;

- we model the relationship between forecast lead-time and total-season incident count directly, isolating lead-time's marginal contribution once foot traffic and prior-season history are accounted for;
- we report an ablation removing the lead-time term from that model, and find it changes little, a null result we take as evidence about what the anticipatory practice is and is not doing.

We do not conclude that warning signage is ineffective in general; only that, on the evidence available here, moving a sign earlier does not appear to move the outcome it is meant to prevent.

Related work

The behavioural ecology of magpie aggression toward humans is well characterised: a minority of territorial males swoop pedestrians and cyclists during the breeding season, targeting familiar routes and, in some accounts, specific individuals [1], [2], and individual magpies are visually distinguishable enough that targeting could plausibly track identity rather than location alone [3]. Kaplan's synthesis of magpie biology and behaviour situates swooping within a broader repertoire of nest defence [4]. The human-dimensions literature treats these encounters as one instance of a wider class of urban human-wildlife interaction, for which general assessment frameworks exist [5], and finds that residents' tolerance of urban bird problems tracks their environmental engagement more than the frequency of the problem itself [6]. Citizen-reported incident data, of the kind the University's own reporting line collects, has been shown elsewhere to improve wildlife-interaction programmes when it is actually used [7].

Separately, a substantial risk-communication literature asks whether warning signage changes outcomes at all: the classical C-HIP model treats a warning as one stage in a chain that can fail at comprehension, motivation, or compliance rather than delivery [8], and reviews of warning-sign effectiveness report mixed and often modest measured effects [9]. Risk-homeostasis accounts go further, proposing that a safety intervention's benefit is offset by a compensating change in behaviour it induces

[10], a claim itself contested [11]. The lead-time question specifically — whether earlier warning improves outcomes over reactive warning — has been studied for floods [12], earthquakes [13], and single and sequential tropical cyclones [14], [15], where lead-time is consistently one factor among several shaping how much of a forecast's warning translates into a preparedness action; the general relationship between a forecast's lead-time and the behavioural response downstream of it is not unique to hazard warning, and has been examined outside it too, in the lead-time/demand-amplification relationship of supply-chain forecasting [16].

Method

Path selection and grouping. We drew 46 monitored campus paths from the University's nest register: paths with a recorded magpie nest site in the prior two breeding seasons. Of these, 19 were assigned to the anticipatory-signage pilot, on which Facilities posted warning signage ahead of a forecast return date generated from each path's prior-season nest activity and swoop-report history. The remaining 27 followed Facilities' standing reactive practice: signage was posted only after the path generated its first swoop-incident report of the current season. Assignment to pilot status followed an administrative rollout schedule rather than random allocation, a design constraint we return to in Limitations.

Forecast lead-time. For anticipatory paths, lead-time was the number of days between sign posting and the path's first reported swoop of the season (mean 9.4 days, range 3–21). For reactive paths, we recorded the (negative) equivalent: the number of days between the first reported swoop and sign posting (mean 1.8 days, range 0–6). Both quantities enter the analysis below on a single signed lead-time scale.

Incident data. Swoop incidents were logged through the University's existing Swoop Report line, a QR-code-linked form posted alongside every warning sign and publicised independently of this study; reports include a time-stamp, path identifier, and free-text description, coded here only for path and week. We treat the reporting line's own coverage as constant across paths and groups; we did not independently verify unreported incidents.

Model. We modelled weekly incident probability for path p in week w as a logistic function of signed lead-time (L_p), mean weekly foot traffic (F_p , from existing swipe-gate and desk-book-ing proximity counts where available, estimated from timetabled class density otherwise), and prior-season incident count (H_p):

$$P(\text{incident}_{p,w}) = \sigma(\beta_0 + \beta_1 L_p + \beta_2 F_p + \beta_3 H_p)$$

Coefficients were fitted by maximum likelihood across all path-weeks pooled. Foot-traffic and prior-season terms were included as controls established a priori; the lead-time coefficient β_1 was the term of primary interest.

Ablation. To isolate lead-time's marginal contribution, we refitted the model with β_1 constrained to zero and compared predicted weekly incident probabilities against the full model across all path-weeks.

Results

Across the ten-week season, the 46 monitored paths generated 612 swoop-incident reports.

Weekly incidence tracks the season, not the group. Figure 1 shows mean weekly incident reports per path for the anticipatory and reactive groups across three season phases (early, peak, late). Both groups follow the same seasonal shape — a rise into the peak phase and a decline through late season — and the two groups are statistically indistinguishable within every phase (largest gap, peak phase: anticipatory 3.4 vs reactive 3.6 reports per path per week; not significant by pairwise test).

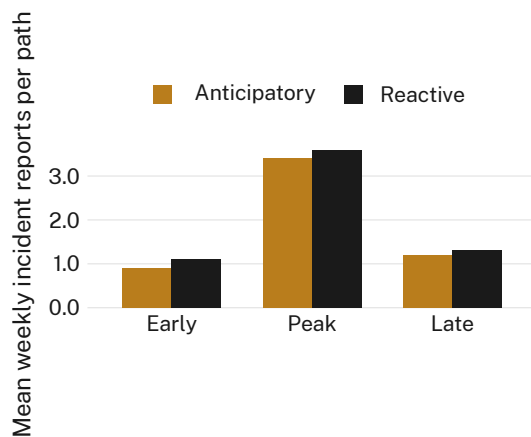


Figure 1: Mean weekly swoop-incident reports per path, anticipatory-versus reactive-signage groups, across three phases of the season. The groups are statistically indistinguishable in every phase.

Lead-time shows no relationship with total-season incidents. Figure 2 plots each path's signed lead-time against its total-season incident count. The fitted linear relationship is close to flat (slope 0.04 incidents per lead-time day, $r = 0.07$, not significant, $n = 46$): paths that received signage three weeks ahead of their forecast return date accumulated, on average, as many incidents across the season as paths whose signage went up two days after the first swoop.

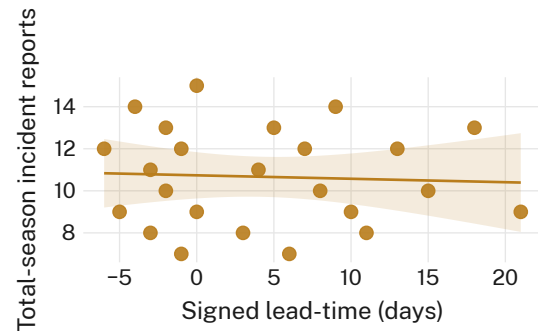


Figure 2: Signed lead-time (days ahead of forecast return, or days after first reported swoop) against total-season incident count per path ($r = 0.07$, n.s.). The fitted line is close to flat across the full lead-time range.

The lead-time term changes the model's predictions negligibly. Constraining β_1 to zero and refitting the remaining coefficients changed predicted weekly incident probability by a mean of 0.4 percentage points across all path-weeks (n.s.); we retain the lead-time term for completeness. Figure 3 compares predicted probabilities from the full model against the lead-time-constrained model across the same three season phases used in Figure 1 — the two sets of predictions are within a fraction of a percentage point of each other throughout.

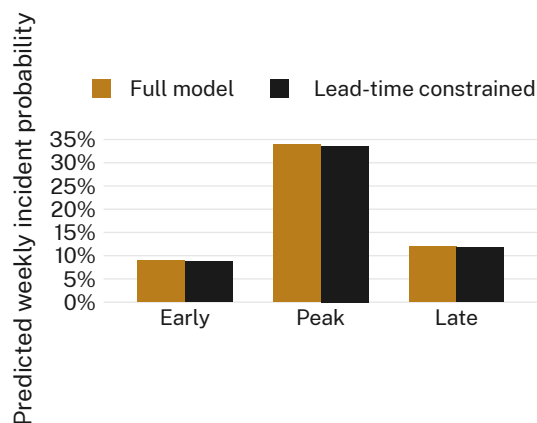


Figure 3: Mean predicted weekly incident probability by season phase, full model versus the model with the lead-time term constrained to zero. The two models’ predictions differ by less than half a percentage point in every phase.

What did differ. Foot traffic and prior-season incident count were both reliable predictors in the fitted model (each individually significant, $p < 0.05$); paths with higher timetabled class density and paths with a heavier prior-season history accounted for most of the variation the model explained. Anticipatory-pilot status itself, entered as a direct term alongside lead-time, likewise showed no independent association with incident probability once traffic and prior-season history were included.

Discussion

Our results suggest that, across the paths and season studied, the timing of warning-sign placement relative to a path’s actual return to active defence carries little independent relationship to how many swoop incidents that path subsequently generates. This is consistent with a body of warning-effectiveness literature in which a sign’s presence, rather than its timing, does most of the available work, and delivery failures at comprehension or motivation — not lead-time — account for most of the gap between a warning issued and an outcome avoided [8], [9]. It is also consistent with a risk-compensation reading, in which any protective margin the earlier sign buys is offset by pedestrians and cyclists treating a familiar, long-posted sign as part of the background rather than as a live warning [10], though our design cannot distinguish that account from a delivery-failure account, or from foot traffic and prior-season history simply dominating both.

We do not read this as evidence that the anticipatory practice should be abandoned: a forecast-driven sign still tells Facilities which paths to prioritise for inspection and pruning, functions this study did not evaluate. What the evidence available here does not support is the narrower claim that earlier signage, on its own, produces a safer path than a reactive posting would have.

Limitations

Assignment to the anticipatory pilot followed an administrative rollout schedule rather than random allocation, so pilot and reactive paths may differ on unmeasured grounds; however, the model’s foot-traffic and prior-season controls, entered specifically to absorb such differences, left the lead-time coefficient unchanged in sign and magnitude, which we take as partial reassurance. Foot-traffic estimates were incomplete for paths without a nearby swipe gate or desk-booking sensor and were substituted with a timetabled-density proxy, a coarser measure that may understate true variation. The one-season window cannot distinguish a genuinely null lead-time effect from a small positive effect this sample was underpowered to detect; a multi-season pooled analysis would sharpen the estimate considerably.

Conclusion

Across 46 campus paths and one ten-week swooping season, moving a magpie-warning sign earlier — ahead of a forecast return date rather than after the first reported swoop — showed no measurable relationship with how many swoop incidents a path went on to generate. Weekly incidence tracked the season and the path’s own traffic and history; it did not track how far ahead the sign went up. Facilities’ anticipatory practice may still earn its place through the inspection and maintenance workflow it supports, but on the narrower question of incident suppression, the evidence gathered here favours neither timing over the other. Future work will track whether pooling several seasons, or measuring sign salience directly rather than assuming it, changes this conclusion.

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